HiRes Processing Pipeline

APPN Aerial SOP — Locked revision 1.00

HiRes Processing Pipeline

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> **Coming soon** — the formal HiRes processing pipeline documentation is still in development. In the meantime, see the Processing Pipelines overview and the HiRes Fieldbook for sensor-side context.

> **Caution**
>
> **AI-generated scaffold — content not validated.** The structure and prose in the sections below were scaffolded using Claude Opus to outline the intended shape of the document. The technical content (toolchain details, step sequences, parameter names, output filenames, etc.) has **not** been reviewed or verified against the actual APPN HiRes processing workflow and should be treated as a placeholder only. Do **not** rely on this content for processing until it has been replaced with verified, operator-authored instructions.

> **Important**
>
> This page documents the **standardised GRYFN processing pipelines** used within APPN for UAV-based data processing, intended for trained APPN staff processing CALViS, GOBI, HiRes, and M3M datasets. Adherence to the pipelines below ensures reproducibility, quality assurance, and cross-project comparability. For any processing run that **deviates from the standard pipeline**, detailed records must be kept of every parameter changed, the rationale, and any anticipated implications for data quality.

This page documents the standardised HiRES (PhaseOne) processing pipeline within APPN for UAV-based data processing. These configurations define consistent, transparent, and repeatable processing workflows across GRYFN-supported sensors and platforms, supporting reproducibility, quality assurance, and cross-project comparability. The files are intended to be used as reference and default templates for approved processing pipelines, with any deviations explicitly documented to maintain traceability and data integrity.

The standard pipelines and the data products they output are detailed below. For a tabular summary of output formats, resolutions and software compatibility, see Standard Data Products.

> **Important**
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> **Document status — work in progress.** This page is a draft and requires further revision before it can be considered final.

Document Structure

- Processing methods
- Method 1 — PhaseOne iX Capture (Windows GUI)
- Method 2 — PhaseOne Image SDK (Linux CLI / shell scripts)
- Method 3 — Display-focused orthomosaic

Processing methods

APPN currently supports **three** processing pathways for HiRes (PhaseOne) data. The first two pathways produce the standard, radiometrically and geometrically rigorous APPN data products and are based on the PhaseOne iX Capture / Image SDK toolchain. The third pathway is a lightweight, display-focused orthomosaic intended for rapid visualisation, QA previews, and stakeholder engagement — it is **not** suitable for quantitative analysis.

#	Method	Toolchain	Operating system	Primary output	Use case
1	iX Capture GUI	PhaseOne iX Capture (GUI)	Windows	Calibrated RGB frames + standard APPN orthomosaic	Default operator-driven processing
2	Image SDK CLI	PhaseOne Image SDK + shell scripts	Linux	Calibrated RGB frames + standard APPN orthomosaic	Batch / HPC / reproducible processing
3	Display orthomosaic	Lightweight photogrammetry (e.g. Metashape / ODM)	Windows or Linux	Visually optimised RGB orthomosaic (GeoTIFF / COG)	Quick-look previews, comms, QA

> **Note**
>
> Methods 1 and 2 are **functionally equivalent** — they apply the same radiometric calibration, lens correction, and geometric model from the PhaseOne Image SDK, and should produce numerically comparable outputs. Method 3 is **separate** and produces a visually pleasing product at the cost of some mosaicing artefacts and lower resolution.

Method 1 — PhaseOne iX Capture (Windows GUI)

> **Coming soon** — full step-by-step instructions to be added.

Toolchain. PhaseOne **iX Capture** (Windows desktop GUI), built on the PhaseOne Image SDK.

When to use. This is the **default** processing pathway for routine operator-driven HiRes processing where a workstation with the iX Capture licence is available.

Inputs.

- Raw PhaseOne `.IIQ` frames from the HiRes sensor.
- Camera calibration / LCC files supplied with the sensor.
- Flight log / event marks (for georeferencing).
- RTK / PPK trajectory (where available).

Outline of steps.

1. Ingest the raw `.IIQ` frames into iX Capture.
2. Apply the supplied PhaseOne **camera calibration** profile.
3. Apply **LCC (Light Calibration / flat-field) correction**.
4. Apply lens distortion correction and geometric model.
5. Export calibrated, georeferenced frames (GeoTIFF / DNG) to be passed to the orthomosaic stage.
6. Generate the standard APPN orthomosaic from the calibrated frames.

Outputs.

- Calibrated per-frame imagery.
- `RGB_Orthomosaic.tif` (GeoTIFF, 0.6 cm fixed GSD — see Standard Data Products).
- Processing report / log.

TODO.

- Document iX Capture version pinned for APPN processing.
- Document exact GUI parameter presets and screenshots.
- Document export settings for handover to the orthomosaic stage.

> **Note**
>
> **Images needed (screenshots):**

- iX Capture GUI showing the camera-calibration profile being loaded / applied.
- LCC (Light Calibration / flat-field) correction step in the GUI.
- Export-settings dialog for calibrated GeoTIFF / DNG frame output.

Method 2 — PhaseOne Image SDK (Linux CLI / shell scripts)

> **Coming soon** — full step-by-step instructions and reference scripts to be added.

Toolchain. PhaseOne **Image SDK** invoked from shell scripts on Linux. This produces outputs that are equivalent to Method 1 but is suitable for **batch processing, headless servers, and HPC**, and is the preferred path for reproducible processing runs.

When to use.

- Large backlogs / batch reprocessing.
- Headless / HPC environments without a Windows GUI.
- Where strict reproducibility (scripted, version-controlled parameters) is required.

Inputs. Same as Method 1.

Outline of steps.

1. Stage raw `.IIQ` frames and calibration files in the expected data folder structure.
2. Run the APPN HiRes shell pipeline, which wraps the PhaseOne Image SDK CLI to:
3. Apply the camera calibration profile.
4. Apply LCC / flat-field correction.
5. Apply lens distortion and geometric correction.
6. Export calibrated GeoTIFF / DNG frames.
7. Pass the calibrated frames to the orthomosaic stage.
8. Emit a machine-readable processing report (parameters, versions, run time, checksums).

Outputs. Same as Method 1, plus a structured processing log.

TODO.

- Document Image SDK version pinned for APPN processing.
- Add reference shell pipeline (location in repo, invocation, env vars).
- Document parity checks against Method 1 (numerical equivalence tests).
- Document container / environment (Conda / Docker) used for runs.

> **Note**
>
> **Images needed (screenshots):**
>
> - Terminal screenshot showing the reference shell-pipeline invocation (env vars, command line, expected log output).
> - Sample structured processing report (parameters, versions, run time, checksums) rendered for the docs.

Method 3 — Display-focused orthomosaic

> **Coming soon** — full step-by-step instructions to be added.

Toolchain. Lightweight photogrammetry suite (e.g. Agisoft Metashape, OpenDroneMap, or equivalent) — **not** the PhaseOne Image SDK.

When to use. Rapid, visually optimised orthomosaics for:

- In-field / same-day quick-look previews.
- Stakeholder communication, reports, and presentation graphics.
- QA of flight coverage and gross georeferencing before committing to the full Method 1 / Method 2 pipeline.

> **Warning**
>
> Method 3 outputs are **display-only**. They are **not** radiometrically calibrated, do **not** carry the standard APPN per-band reflectance properties, and **must not** be used for quantitative analysis, reflectance retrieval, or downstream spectral / trait modelling.

Inputs.

- Raw or quickly-developed RGB JPEGs / TIFFs from the HiRes frames.
- Flight log / image GPS tags.

Outline of steps.

1. Develop / export RGB previews from the raw `.IIQ` frames (default profile, no rigorous calibration required).
2. Load frames into the chosen photogrammetry tool.
3. Run alignment, dense cloud / depth maps (optional), and orthomosaic generation with display-oriented presets (visual colour balancing permitted).
4. Export as GeoTIFF / Cloud Optimized GeoTIFF for web / GIS preview.

Outputs.

- `RGB_Display_Orthomosaic.tif` (GeoTIFF / COG, visually optimised, uncalibrated). Filename and folder convention to be confirmed.

TODO.

- Confirm preferred photogrammetry tool and version.
- Document filename / folder convention so display products are clearly distinguishable from the standard `RGB_Orthomosaic.tif`.
- Document required watermark / metadata flag marking the product as "display only — not for quantitative use".